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SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Final version — Activities 1.1, 1.2 and 1.3

Nama Nameng: A Mobile and Web-Based Ordering, Payment and Queue Management System for Nama Nameng

External Stakeholder: Nama Nameng (Chisa Nyama)

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Group Members

#	Full Name	Student Number
1	Neo Lekhelebana	221000484
2	Charmaine Mogotsi	221023182
3	Portia Nkotseng Sefatsa	220007164
4	Tshepang Morake	222075924
5	Hlayiseka Matsimbi	221027400

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1. Cover Page and Group Members

The cover page of this document identifies the institution, module, project title, external stakeholder and the group members responsible for this Work-Integrated Learning (WIL) project. Full names, student numbers and individual roles of all group members are recorded in the table on the cover page.

2. Project Title

Nama Nameng: A Mobile and Web-Based Ordering, Payment and Queue Management System for Nama Nameng

The project title reflects the core purpose of the system: connecting customers digitally to a chisa nyama business that currently relies entirely on in-person, walk-up ordering.

3. Problem Statement

3.1 Stakeholder and Organisation

Nama Nameng is a local chisa nyama that prepares and sells grilled meat and accompanying sides to walk-in customers in the surrounding community.

3.2 Current Situation

Nama Nameng currently operates using a fully manual, in-person process. Customers must physically travel to the premises, join a queue, and wait to reach the counter before they can even place an order. There is no app or web-based ordering option available, they only take orders using owner's personal WhatsApp. The business has no website, no online menu, and no digital presence beyond informal word-of-mouth recommendations from existing customers.

3.3 The Problem

- Customers experience long, visible physical queues, especially during peak trading times (weekends, lunch hours and paydays).
- There is no way for a customer to place an order in advance, browse the menu remotely, or check what is currently available before arriving.
- The business has no digital record of orders, sales trends, or customer preferences, making it difficult to plan stock or staffing.
- Marketing relies entirely on word-of-mouth, which limits the business's ability to reach new customers outside its immediate local network.
- Order-taking is manual and verbal, which increases the risk of miscommunication, forgotten items and incorrect orders.

3.4 Affected Users

- Walk-in and repeat customers of Nama Nameng, who lose time standing in queues.

- The owner/manager of Nama Nameng, who cannot track orders, sales or customer demand digitally.
- Kitchen/braai staff, who receive orders verbally with no queue or ticketing system.
- Potential new customers in the surrounding area who are unaware of the business due to its reliance on word-of-mouth marketing.

3.5 Consequences and Need for Improvement

If the current manual process continues unchanged, Nama Nameng risks losing customers to competitors who offer more convenient ordering options, continues to miss opportunities to grow its customer base beyond its immediate community, and remains unable to make data-informed decisions about its menu, stock and staffing. A digital ordering and queue management solution is needed to reduce queuing time, extend the business's reach, and provide the owner with visibility over orders and sales.

4. Purpose, Aim and Objectives

4.1 Purpose

The purpose of this project is to design and develop a digital ordering and queue management solution for Nama Nameng, replacing the current fully manual, word-of-mouth-driven process with a mobile application for customers and a web-based management dashboard for the business, supported by a shared cloud backend.

4.2 Aim

To reduce customer queuing time and improve order accuracy at Nama Nameng by enabling customers to browse the menu and place orders digitally, while giving the business owner a simple, reliable way to manage orders, menu items and basic sales information online.

4.3 Objectives

- To develop a cross-platform mobile application (Flutter) and a web-based management (ASP.NET) that allows customers to view the menu, place and pay for orders, and track order status remotely which also allows Owner/staff to manage the menu, view and update incoming orders, and view basic order/sales history.
- To implement a shared backend (Supabase) that stores menu, order and user data, and synchronises data in real time between the mobile app and the web dashboard.
- To reduce physical queuing by allowing customers to place orders ahead of arrival and receive a notification or status update when their order is ready.
- To provide the business with a basic digital record of orders that can support future marketing and stock-planning decisions.

5. Project Scope

This initial SRS defines the scope of the Nama Nameng system as it will be developed for the WIL project prototype. The scope focuses on ordering, menu management and queue-visibility functionality that directly addresses the problems identified with Nama Nameng.

5.1 In Scope

- To develop a cross-platform mobile application (Flutter) and a web-based management (ASP.NET) that allows customers to view the menu, place and pay for orders, and track order status remotely which also allows Owner/staff to manage the menu, view and update incoming orders, and view basic order/sales history.
- A shared Supabase backend providing the database, authentication and real-time synchronisation between the mobile app and web dashboard.
- Basic digital queue/ticket visibility so customers know their order position or estimated wait.
- Basic order history and simple sales overview for the owner.

6. Deliverables and Acceptance Criteria

Deliverable	Acceptance Criteria	Owner
Flutter mobile app & ASP.NET web app	To develop a cross-platform mobile application (Flutter) and a web-based management (ASP.NET) that allows customers to view the menu, place and pay for orders, and track order status remotely which also allows Owner/staff to manage the menu, view and update incoming orders, and view basic order/sales history.	Group
Supabase backend (database & auth)	All order, menu and user data is stored centrally, and changes made from either app sync correctly and promptly across both platforms.	Group
Digital menu	Menu items, descriptions and prices are accurate, current, and reflect what Nama Nameng actually offers, as confirmed by the stakeholder.	Group
Order status / queue visibility	Customer can see their order's current status (e.g. Received / Preparing / Ready) after placing it.	Group
Initial and final SRS documentation	SRS accurately reflects stakeholder-confirmed requirements and is approved/acknowledged by the stakeholder.	Group

7. Exclusions, Constraints and Assumptions

7.1 Exclusions

- Integration with third-party delivery platforms (e.g. Uber Eats, Mr D Food) is excluded from this initial prototype.
- Integrated card/online payment gateway processing is excluded from the initial prototype; payment recording will be simulated or limited to cash-on-collection/manual confirmation unless the stakeholder confirms otherwise.
- Loyalty/rewards programmes, advanced analytics and multi-branch support are excluded from this version.
- Native iOS-specific and Android-specific features beyond what Flutter provides by default are excluded.

7.2 Constraints

- The project must be completed within the academic semester timeframe set by the ITC327W module.
- Development is limited to the technology stack specified by the module: Flutter (mobile), ASP.NET (web), and Supabase (backend/database).
- Access to the stakeholder for requirement confirmation is limited to arranged visits/meetings and is dependent on the stakeholder's business operating hours.

7.3 Assumptions

- Nama Nameng has, or is willing to obtain, basic internet connectivity (Wi-Fi/mobile data) at the premises to run the staff web dashboard.
- Customers placing orders through the mobile app have access to a smartphone and mobile data.
- The stakeholder will remain available for periodic feedback and requirement confirmation throughout the project.
- The menu and pricing information provided by the stakeholder during initial engagement is accurate and will be confirmed before final development.

8. Stakeholders and User Groups

Stakeholder / User Group	Description	Interest / Involvement
Nama Nameng (Owner/Manager)	Primary external stakeholder; owns and operates the chisa nyama business.	Provides requirements, confirms scope, approves menu content and system behaviour.
Kitchen/Braai Staff	Prepares food and currently receives orders verbally.	Will use the order-status/queue features of the web & app dashboard.
Customers (Walk-in & Repeat)	Community members who purchase food from Nama Nameng.	End users of the Flutter mobile app & ASP.NET web app, primary beneficiaries of reduced queuing.
Project Group (Students)	WIL project team responsible for design, development and delivery.	Gathers requirements, designs and builds the Flutter, ASP.NET and Supabase system.
ITC327W Module Facilitator	Assessor of the WIL project.	Reviews and assesses project deliverables, including this SRS.

Stakeholder needs

The following needs were identified directly through engagement with Doja (Nama Nameng owner) during the initial stakeholder meeting (see Appendix A) and confirmed as accurate for this project (see Appendix B). These needs directly inform the functional and non-functional requirements in Sections 12 and 13.

- Reduce the time customers spend standing in physical queues, particularly during peak periods (weekends, lunch hours, paydays).
- Provide a way for customers to see the menu and place an order before or without needing to be physically present at the counter.
- Reduce order-taking errors caused by verbal, in-person communication between customers, counter staff and the kitchen/braai area.
- Give the owner a simple way to manage the menu (add, edit, remove items and prices) without needing technical assistance.
- Give the owner visibility of incoming orders and the ability to update order status so customers know when their food is ready.
- Keep the solution simple and low-cost to operate, given the business's current reliance on informal, low-tech processes (WhatsApp, verbal ordering) and limited digital literacy among staff.
- Retain flexibility around payment — the business currently operates on a cash/in-person payment basis, and a full online payment gateway is not a priority need at this stage.
- Build a basic digital record of orders over time, to support future decisions on stock and staffing (a need the business cannot currently meet at all).

These needs were used to validate that the functional requirements in Section 12 solve real, confirmed problems rather than assumptions made by the group.

9. Current Process and Diagram

Nama Nameng's current ordering process is entirely manual and in-person, as illustrated in Figure 1 below.

9.1 Current Process Description

- A customer physically arrives at the premises; there is no remote ordering option.
- The customer joins a physical queue and waits their turn to reach the counter.
- Once at the counter, the customer verbally tells a staff member what they would like to order.
- The staff member relays the order to the kitchen/braai area verbally, with no ticket or digital queue reference.
- Food is prepared with no visibility to the customer of their position in the preparation queue.
- The customer pays at the counter (cash or card) and waits for the food to be ready.
- The customer receives their food and leaves; if satisfied (or not), they may tell others about their experience — this word-of-mouth sharing is currently the business's only form of marketing.

Figure 1: Current (Manual) Ordering Process at Nama Nameng

Word-of-mouth is the ONLY marketing channel: satisfied/unsatisfied customers tell others; no digital record, no online reach, no repeat-order data



Pain points: long visible queues, no way to order ahead, no order tracking, no digital customer reach, no sales/order data captured, business growth limited to local word-of-mouth reputation.

9.2 Key Pain Points Identified

- Long, visible physical queues, particularly at peak times.
- No way to check the menu remotely.
- No digital record of orders, sales or customer information.
- Marketing and customer growth limited entirely to word-of-mouth.

10. Product Perspective

Nama Nameng is a new, self-contained digital system developed specifically to replace Nama Nameng's manual ordering process. It is not a modification of an existing system.

The system consists of these integrated components:

- A Flutter-based mobile application & an ASP.NET web app used by customers to browse the menu, place orders and track order status. Staff use the aforementioned flutter and ASP.NET web-app to manage the menu and process incoming orders.
- A shared Supabase backend that provides the database, authentication and real-time data synchronisation connecting both front-ends.

The system will operate independently of any existing third-party ordering platforms and will be tailored specifically to Nama Nameng's menu, operating hours and day-to-day workflow, as confirmed through stakeholder engagement.

11. Main Flutter, ASP.NET and Supabase Functions

11.1 Flutter Mobile Application & ASP.NET Web (Customer-Facing)

- Customer registration and login.
- Browse digital menu with item descriptions, categories and prices.
- Add items to a cart and place an order.
- View real-time order status (e.g. Received, Preparing, Ready for Collection).

- View basic order history for the logged-in customer.
- Receive in-app notifications/status updates on order progress.

11.2 1 Flutter Mobile Application & ASP.NET Web Dashboard (Owner/Staff-Facing)

- Staff/owner authentication and secure login.
- Add, edit, and remove menu items, categories and prices.
- View incoming orders in real time as they are placed.
- Update order status so customers see live progress in the mobile app.
- View a basic list/history of past orders and simple daily sales overview.

11.3 Supabase Backend

- Central relational database storing users, menu items, orders and order status.
- Authentication service for both customer & staff using (Flutter & ASP.NET) accounts.
- Real-time subscriptions so order status changes made on the web dashboard reflect instantly in the mobile app, and new orders reflect instantly on the dashboard.
- Storage for menu item images.
- Row-level security rules to ensure customers can only view/edit their own orders and staff-only data is protected.

12. Functional Requirements

ID	Requirement Description	Linked Stakeholder Need	Priority
FR-01	The system shall allow a customer to register and log into the mobile & web app.	Remote/self-service access	High
FR-02	The system shall allow a customer to view the current digital menu with prices.	Browse menu before arriving	High
FR-03	The system shall allow a customer to add items to a cart and submit an order.	Order ahead of arrival	High
FR-04	The system shall allow a customer to view the real-time status of their order.	Reduce uncertainty/queue time	High
FR-05	The system shall allow staff/customers to log into the web & app dashboard securely.	Business data security	High
FR-06	The system shall allow staff/owner to add, edit and remove menu items and prices.	Keep menu accurate/current	High
FR-07	The system shall allow staff/owner to view all incoming orders in real time.	Reduce miscommunication	High
FR-08	The system shall allow staff/owner to update the status of an order.	Inform customer of progress	High
FR-09	The system shall notify/update the customer when their order status changes.	Reduce physical waiting	Medium

ID	Requirement Description	Linked Stakeholder Need	Priority
FR-10	The system shall store completed order history accessible to the customer.	Reorder convenience	Medium
FR-11	The system shall provide the owner with a basic overview/list of past orders.	Business insight/planning	Medium
FR-12	The system shall synchronise order and menu data in real time between the mobile app and the web dashboard via Supabase.	Consistent, accurate information	High

13. Non-Functional Requirements

Category	Requirement	Description
Usability	NFR-01	The mobile app shall be simple and intuitive enough for first-time customers with limited technical experience to place an order without assistance.
Performance	NFR-02	The system shall reflect a new order on the staff dashboard within a few seconds of the customer submitting it.
Security	NFR-03	Customer and staff accounts shall be protected using secure authentication, and customers shall only be able to access their own order data.
Availability	NFR-04	The system shall be available during Nama Nameng's normal trading hours, with minimal planned downtime.
Reliability	NFR-05	The system shall accurately save and display order and menu data without data loss under normal operating conditions.
Scalability	NFR-06	The system shall be able to support a growing number of customers and orders as the business's digital reach expands, without major redesign.
Maintainability	NFR-07	The codebase shall be structured and documented so that future developers/students can understand, maintain and extend it.
Compatibility	NFR-08	The Flutter mobile app shall function on common Android and iOS smartphone devices used by the target customer base.

14. System Requirements

14.1 Client-Side (Mobile - Customer)

- Smartphone running Android 8.0+ or iOS 13+.
- Active internet connection (Wi-Fi or mobile data).

14.2 Client-Side (Web - Staff/Owner)

- A device (PC, laptop or tablet) with a modern web browser (Chrome, Edge, Firefox or Safari).

- Active internet connection at the Nama Nameng premises.

14.3 Development Environment

- Flutter SDK and Dart for mobile application development.
- .NET SDK (ASP.NET Core) for the web dashboard.
- Supabase account/project for the backend database, authentication and real-time services.
- Git and a GitHub repository for source control and collaboration.
- Code editor Visual Studio Code & Visual Studio.

14.4 Server/Backend Requirements

- Cloud-hosted Supabase project (PostgreSQL database, authentication, storage and real-time services).
- Sufficient storage for menu images and order data within Supabase's free/available tier for the prototype.

15. Integration Requirements

The Flutter mobile application, ASP.NET web dashboard and Supabase backend must operate as a single, integrated solution rather than three separate systems.

- The Flutter app shall communicate with Supabase directly (via the Supabase client SDK) for authentication, data reads/writes and real-time order updates.
- The ASP.NET web dashboard shall communicate with Supabase (via REST/API calls or the appropriate .NET client) to manage menu data and retrieve/update orders.
- Both the Flutter app and ASP.NET dashboard shall share the same underlying Supabase database schema for users, menu items and orders, ensuring a single source of truth.
- Real-time subscriptions shall be used so that an order placed on the Flutter app appears immediately on the ASP.NET dashboard, and a status update made on the dashboard is reflected immediately in the Flutter app.
- Authentication tokens issued by Supabase shall be used consistently across both the mobile and web applications to secure access to data.

16. Requirements Prioritisation

The MoSCoW method was used to prioritise the functional requirements listed in Section 12, based on stakeholder input on what is essential to solving the core problem versus what would add value later.

Priority	Requirements
Must Have	FR-01, FR-02, FR-03, FR-04, FR-05, FR-06, FR-07, FR-08, FR-12 — core registration, ordering, menu management, order visibility and real-time synchronisation.
Should Have	FR-09, FR-10, FR-11 — order notifications, order history and basic sales overview, which improve the experience but are not strictly required for the core problem to be solved.

Priority	Requirements
Could Have	Digital queue position/estimated wait time indicator; ratings/reviews on completed orders.
Won't Have (this version)	Third-party delivery integration, online payment gateway processing, loyalty/rewards programme, multi-branch support (see Section 7.1 Exclusions).

Requirements Change Log

The table below records notable changes made to the requirements and scope after the feasibility study and risk analysis were completed. These example entries reflect the type of change typically required after feasibility/risk review — replace the details with the actual changes your group agreed on.

Change ID	Requirement Affected	Change Description	Reason for Change	Date
CL-01	FR-04 / NFR-02	Order status update target relaxed from "instant" to "within a few seconds".	Technical feasibility review found Supabase real-time sync on the free tier is fast but not guaranteed to be instantaneous; a few-second target is realistic and still solves the customer's core need.	26 August 2026
CL-02	Section 7.1 Exclusions	Online/card payment gateway integration confirmed as fully excluded from this prototype (previously a possible "maybe").	Economic and schedule feasibility analysis showed payment gateway integration would add cost, compliance and testing overhead the group cannot realistically absorb this semester.	27 August 2026
CL-03	FR-09 (notifications)	In-app status notification confirmed as sufficient; push/SMS notifications explicitly excluded.	Schedule feasibility — push/SMS notification services would require additional third-party integration and testing time not available in the current schedule.	28 August 2026
CL-04	Section 5.1 Scope	Digital queue position/estimated wait time moved from "in scope" to "Could Have" (Section 16).	Risk analysis (see Risk Register, R-03) identified this as a higher-complexity feature that could threaten schedule feasibility if treated as mandatory.	29 August 2026

Refined Project Scope

Following the feasibility study and risk analysis, the scope defined in Section 5 has been reviewed and refined into the realistic baseline that will be taken forward into system design. This refined scope reflects the changes recorded in the Requirements Change Log above.

Confirmed In Scope (Baseline for Design)

- Customer registration and login (Flutter mobile app).
- Digital menu browsing with categories, descriptions and prices.
- Order placement (cart and submit) from the mobile app.
- Order status visibility (Received / Preparing / Ready) updated within a few seconds via Supabase real-time sync.
- Owner/staff menu management (add, edit, remove items and prices) via the ASP.NET web dashboard.

- Owner/staff order management — view incoming orders and update order status.
- Basic order history for customers and a simple sales overview for the owner.
- Shared Supabase backend providing the database, authentication and real-time synchronisation.

Confirmed Out of Scope (This Version)

- Online/card payment gateway integration — payment remains cash-on-collection/manual confirmation.
- Third-party delivery platform integration.
- Push/SMS notifications — in-app status updates only.
- Loyalty/rewards programme, advanced analytics, and multi-branch support.

Deferred (Could Have ,Time Permitting)

- Digital queue position / estimated wait time indicator.
- Ratings or reviews on completed orders.

This refined scope is considered the realistic, agreed baseline for system design and will only be changed again through a documented update to the Requirements Change Log above.

Feasibility Study

The proposed Nama Nameng system was evaluated across six feasibility areas to confirm the project is realistic before system design begins.

Technical Feasibility

The group has working knowledge of Flutter, ASP.NET and Supabase from module coursework, and all three technologies are freely available with strong documentation and community support. The main technical challenge is real-time synchronisation across three separate components (mobile app, web dashboard and backend); this is achievable using Supabase's built-in real-time subscription features, but should be tested early through a small technical prototype/spike before full development begins, to confirm response times and reliability on the free-tier plan. Conclusion: technically feasible, provided real-time integration is prototyped and validated early.

Operational Feasibility

The solution directly addresses confirmed pain points (queuing time, order miscommunication, lack of digital records) and the stakeholder has expressed willingness to adopt it. Successful operation depends on the owner and staff being comfortable using the web dashboard, and on customers having access to a smartphone and mobile data — both reasonable assumptions for this customer base but worth revisiting with the stakeholder before final rollout. Conclusion: operationally feasible, provided basic orientation/training is given to the owner and staff once the dashboard is ready.

Economic Feasibility

As a student WIL prototype, there is no formal budget. Development tools (Flutter SDK, .NET SDK, VS Code/Visual Studio, Git/GitHub) are free, and Supabase's free tier is sufficient for a prototype of this scale (menu items, orders and basic user accounts). The main resource cost is the group's time. Conclusion: economically feasible within the constraints of a student prototype, with no material cost to the stakeholder.

Schedule Feasibility

The project must be completed within the ITC327W semester timeframe, which is a tight but workable constraint given the scope defined above. Applying MoSCoW prioritisation (Section 16) — building all "Must Have" requirements first — gives the group a realistic path to a working core system, with "Should Have" and "Could Have" items only attempted if time remains. This is the highest-rated risk in the Risk Register (R-07) and must be actively monitored. Conclusion: feasible provided the Must Have requirements are prioritised and the Microsoft Project schedule is followed closely, with Should/Could Have items descope if the team falls behind.

Legal Feasibility

The system will store customer and order data, which must be handled responsibly. Supabase row-level security (NFR-03) will be used to ensure customers can only access their own data, and the group will avoid collecting any personal information beyond what is required for the system to function (e.g. no ID numbers or unnecessary personal details). No passwords, API keys, or confidential stakeholder information will be uploaded to the public GitHub repository, in line with the module's instructions. Conclusion: legally feasible, provided data minimisation and access-control practices are followed throughout development.

Ethical Feasibility

The stakeholder is participating voluntarily and has been informed of the project's purpose as a student learning exercise, not a guaranteed production system. The group will continue to represent the prototype's capabilities honestly to the stakeholder, respect the business's operating hours when requesting meetings, and only capture stakeholder evidence that is necessary and relevant (Section 17 note). Conclusion: ethically feasible, provided the group maintains transparent, respectful and honest engagement with the stakeholder throughout the remainder of the project.

Overall Feasibility Decision

The Nama Nameng project is considered FEASIBLE WITH CHANGES. The core prototype (customer ordering, staff menu/order management, and real-time synchronisation) is realistic within the semester timeframe and the group's current skills, provided: (1) real-time integration between Flutter, ASP.NET and Supabase is prototyped early to confirm technical feasibility, and (2) the group strictly follows MoSCoW prioritisation to protect schedule feasibility, descope Should/Could Have features if necessary. These conditions are reflected in the Refined Project Scope above and in the Risk Register below.

Project Risk Register

The following risks were identified as realistic threats to the Nama Nameng project, using a 1–5 scale for Likelihood and Impact (Low = 1–2, Medium = 3, High = 4–5). Risk Score = Likelihood × Impact; scores of 15 and above are rated High.

ID	Risk Statement	Category	Likelihood	Impact	Score	Level	Mitigation	Contingency
R-01	Doja (stakeholder) has limited availability due to business operating hours, delaying requirement confirmation.	Stakeholder	Medium (3)	Medium (3)	9	Medium	Schedule meetings in advance around trading hours; use WhatsApp for quick confirmations where a full meeting isn't possible.	Proceed with best available information; confirm formally at next available opportunity.
R-02	A group member becomes unavailable or under-contributes, delaying assigned tasks.	Team / Resource	Medium (3)	High (5)	15	High	Track tasks and progress weekly via the Progress Tracker and GitHub activity; hold short weekly check-ins.	Redistribute the tasks among remaining members; escalate to module facilitator if necessary.

R-03	Real-time integration between Flutter, ASP.NET and Supabase proves more complex than expected.	Technical	Medium (3)	High (5)	15	High	Build a small technical prototype/spike of the real-time sync early, before full feature development begins.	Fall back to periodic polling/refresh if true real-time subscriptions prove unreliable.
R-04	Group members have limited prior experience with Supabase and/or ASP.NET, slowing development.	Technical / Skills	Medium (3)	Medium (3)	9	Medium	Allocate dedicated learning time early; use pair programming and official documentation/tutorials.	Seek guidance from module facilitator to simplify the affected feature's implementation if needed.
R-05	Stakeholder requests additional features beyond the agreed scope (e.g. delivery, loyalty programme).	Scope	Medium (3)	Medium (3)	9	Medium	Clearly communicate confirmed exclusions (Section 7.1); confirm scope in writing (Appendix B).	Log the request in Requirements C. Log as a future enhancement, not this version.
R-06	Unreliable Wi-Fi/data connectivity at the Nama Nameng premises affects web dashboard usability.	Operational / Technical	Low (2)	Medium (3)	6	Medium	Confirm connectivity during site visits; test the dashboard using mobile data as a backup.	Investigate a simple offline/cached version of recent orders if connectivity proves unreliable.
R-07	The academic semester timeframe is not enough to complete all planned features to a working standard.	Schedule	High (5)	High (5)	25	High	Follow MoSCoW prioritisation strictly — build all Must Have requirements first; track progress against Microsoft Project weekly.	Descope Should Have requirements if the group falls behind schedule.
R-08	Customer or stakeholder data is mishandled, stored insecurely, or exposed publicly.	Legal / Ethical	Low (2)	High (5)	10	Medium	Apply Supabase row-level security; avoid collecting unnecessary personal data; never upload credentials/API keys to GitHub.	Review the data handling approach with the module facilitator if uncertainty arises.
R-09	Code or documentation is lost or overwritten due to poor version control practices.	Technical / Process	Low (2)	Medium (3)	6	Medium	Commit to GitHub regularly with clear messages; use branches for major features.	Maintain periodic local/cloud backups of key documents.
R-10	Miscommunication within the group leads to duplicated or conflicting work.	Team / Process	Medium (3)	Medium (3)	9	Medium	Weekly check-ins; shared Progress Tracker; clear task ownership recorded in Microsoft Project.	Project Lead manages tasks and reassigns ownership as soon as identified.

Three Highest-Rated Risks

- R-07 — Schedule constraint: the semester timeframe may not be enough to complete all planned features (Score: 25).
- R-02 — Team member unavailability or under-contribution could delay assigned tasks (Score: 15).
- R-03 — Real-time integration between Flutter, ASP.NET and Supabase proving more complex than expected (Score: 15).

These three risks will be actively monitored throughout the remainder of the project and reviewed at each stakeholder/group check-in.

SWOT Analysis — Nama Nameng Ordering & Queue Management System

	Helps	Hurts
Internal	Strengths	Weaknesses
	• Solves real, stakeholder-confirmed pain points (queuing time, order-taking errors) — not assumptions • Team already has working knowledge of Flutter, ASP.NET and Supabase • Entirely free-tier tools — no cost to the stakeholder or the team • Supabase real-time sync gives instant visibility between mobile app and web dashboard • MoSCoW prioritisation protects the core deliverable if time runs short	• Real-time sync across three separate components (Flutter, ASP.NET, Supabase) is technically complex and unproven at the time of writing • No online payment gateway in this version — still relies on cash/manual confirmation • Very tight semester timeframe, with almost no schedule slack (this is your highest-rated risk, R-07) • Access to the stakeholder is limited to Nama Nameng's trading hours • It's a prototype — no support or maintenance plan once the module ends
External	Opportunities	Threats
	• A basic digital order/sales record opens the door to future stock and staffing decisions the business can't make today • Extends the business's reach beyond word-of-mouth to new customers • Clear path to add loyalty programmes, analytics, or notifications in a later phase • If it works, the same model could be adapted for other small food vendors in similar situations	• Team member unavailability or under-contribution could delay tasks (R-02) • Real-time integration proving harder than expected could eat into schedule (R-03) • Customer adoption assumes smartphone + mobile data access, which hasn't been fully confirmed • Supabase free-tier limits could be hit if usage grows beyond prototype scale • A competitor could adopt similar digital ordering faster and capture the "convenience" advantage first

17. Appendices: Stakeholder Evidence

This section will contain the evidence gathered during stakeholder engagement for Activities 1.1–1.3, confirming that the problem statement, requirements and scope above are based on actual stakeholder input rather than group assumptions, as required by the module.

Appendix A: Interview Notes

Stakeholder Requirements Questionnaire

Stakeholder / Business Name: Nama Nameng Unswagana

Respondent Name & Role: Doja

Date of Engagement: 26/08/2024

Conducted by (agent number name(s)): Tshepang Morake

Q1. How long has Nama Nameng been operating, and what are your usual trading days and hours?
8 months, 9am - 10pm

Q2. On a typical day, roughly how many customers do you serve? Does this change a lot on weekends, month-end/paydays, or holidays?
It depends whether its weekend/payday but usually more on night end

Q3. How do customers currently place their order (in person, phone, what)? Is there ever a way to order ahead of time?
In person and via my personal whatsapp

Q4. How do you currently handle payment (cash, card, other)?
☐ Cash only ☐ Card only ☒ Both cash and card ☐ Other:

Q5. What are the biggest frustrations you or your staff experience with the current (manual) way of taking orders?
forgetting orders and irated customers

Q6. Do customers ever complain about waiting times or queues? How do they usually react?
yes, lay (wait) and other leave

Q7. Have there been instances with orders (e.g. wrong item, forgotten item) because orders are taken verbally? How often does this happen?
yes, maybe once a week

Q8. How do new customers currently find out about Nama Nameng?
☒ Word of mouth ☐ Social media ☐ Walking past ☐ Other:

Q9. If customers could order ahead of time using a mobile app, how do you think that would change your day-to-day operations?
It will decrease long queue waiting order order

Q10. How would you like to be notified of a new order (e.g. on screen alert, sound, notification)?
Sound refresh and screen alert

Q11. Would you want customers to be able to track their order status (e.g. Received, Preparing, Ready) from their phone?
☒ Yes ☐ No ☐ Not sure

Q12. Would you be interested in customers being able to pay through the app, or would you prefer payment to take place in cash/card on collection for now?
☐ Pay in app ☒ Keep payment on collection ☐ Not sure yet

Stakeholder Confirmation

I confirm that the responses recorded in this questionnaire accurately reflect my views on the current process and needs of Nama Nameng, and that this information may be used by the student project group to inform their Software Requirements Specification.

Stakeholder Signature: Doja Date: 26/08/2024

Email Address: doja@nama-nameng.co.za Phone Number: 063 706 3353

Approved by (Project Manager): Tshepang Morake Date: 26/08/2024

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Appendix B: Requirements Confirmation Evidence

CENTRAL UNIVERSITY OF TECHNOLOGY, FREE STATE
Faculty of Engineering, Built Environment and Information Technology
WORK-INTEGRATED LEARNING – Module Code: ITC327W

Meeting Minutes

Nama Nameng: Mobile and Web-Based Ordering, Payment and Queue Management System

Meeting Purpose: Initial Stakeholder Meeting

Date: 18 August 2025

Time: [Insert start – end time]

Venue / Platform: Nama Nameng premises

Minutes Recorded By: Portia Nkotseng Sefatsa

1. Attendees

Full Name	Student Number	Role / Organisation
Neo Lekhelebane	22100404	Project Group
Charmaine Mogotsi	221023182	Project Group
Portia Nkotseng Sefatsa	220007164	Project Group
Tshepang Morake	222075924	Project Group
Hlayiseka Matsimbi	221027400	Project Group
Doja	N/A	Nama Nameng (External Stakeholder)

2. Apologies / Absent

None

3. Agenda and Discussion Summary

Item	Discussion Summary	Action / Outcome
1	The group introduced themselves and explained the purpose of the WIL project (ITC327W) to design and develop a mobile ordering app and web-based management dashboard for Nama Nameng as part of the group's Diploma in Information Technology coursework.	Doja welcomed the group and agreed to participate as the project's external stakeholder.
2	The group explained that the system will consist of a Flutter mobile app and ASP.NET web app for customers owner/staff, and a shared Supabase backend, and that this initial meeting was to understand the current business process before requirements are finalised.	Doja confirmed understanding of the project and agreed to describe how the business currently operates.
3	Doja explained that Nama Nameng currently operates on a fully manual, in-person basis: customers must physically visit the premises, join a queue, and order verbally at the counter. Some orders are also taken informally through the owner's personal WhatsApp. There is no website, no online menu, and no formal record-keeping system.	Confirmed as the current process to be documented in Section 9 (Current Process and Diagram) of the SRS.
4	Doja confirmed that queues become long and visible during peak periods such as weekends, lunch hours and paydays, and that verbal order-takingsometimes leads to miscommunication, forgotten items or incorrect orders. Doja also noted there is currently no way to track sales trends or plan stock digitally, and	Pain points confirmed and aligned with Section 3.3 (The Problem) of the SRS – no changes required.

that marketing relies entirely on word-of-mouth.

5. The group presented the proposed solution: a mobile app allowing customers to browse the menu, place and pay for orders remotely, and track order status (Received / Preparing / Ready), and a web dashboard allowing Doja to manage the menu, view/update incoming orders, and view basic sales history. Doja provided the current menu items, descriptions and prices to be used in the app.

6. The group clarified that, for this initial prototype, integration with third-party delivery platforms, online/card payment gateways, and loyalty or multi-branch features are out of scope, and that payment will be simulated or handled as cash-on-collection. The group also confirmed that Nama Nameng has basic Wi-Fi/data connectivity available at the premises to run the web dashboard.

7. The group explained that an initial SRS would be compiled based on this meeting and would be brought back to Doja for review and confirmation, along with a request for written WhatsApp confirmation of the menu and scope, and permission to take site photographs for documentation purposes.

Doja confirmed the proposed solution address the business's needs. Menu list and prices to be documented in Section 11 and used as evidence for Appendix B.

Doja agreed to these exclusions and constraints for the initial version of the system.

Doja agreed to review the SRS once compiled and to provide confirmation via WhatsApp.

4. Key Decisions / Confirmations

- Doja confirmed the current menu items, descriptions and prices as accurate for use in the SRS and mobile/web app.
- Doja confirmed the in-scope features: menu browsing, order placement, order status/queue visibility, and basic sales history for the owner.
- Doja agreed that online/card payment gateway integration, delivery platform integration and loyalty features are excluded from this initial version.
- Doja confirmed Wi-Fi/data connectivity is available at the premises for the web dashboard.

5. Action Items

Action Item	Responsible	Due Date
Compile the initial SRS based on this meeting's All Group Members findings	Portia Nkotseng Sefatsa, Charmaine Mogotsi	25 August 2026
Obtain WhatsApp confirmation of menu items, prices and scope from Doja	Neo Lekhelebane, Tshepang Morake	22 August 2026
Take site photographs of current ordering/queue setup (with permission)		23 August 2026

6. Next Meeting

Date: [Insert date of next planned follow-up]

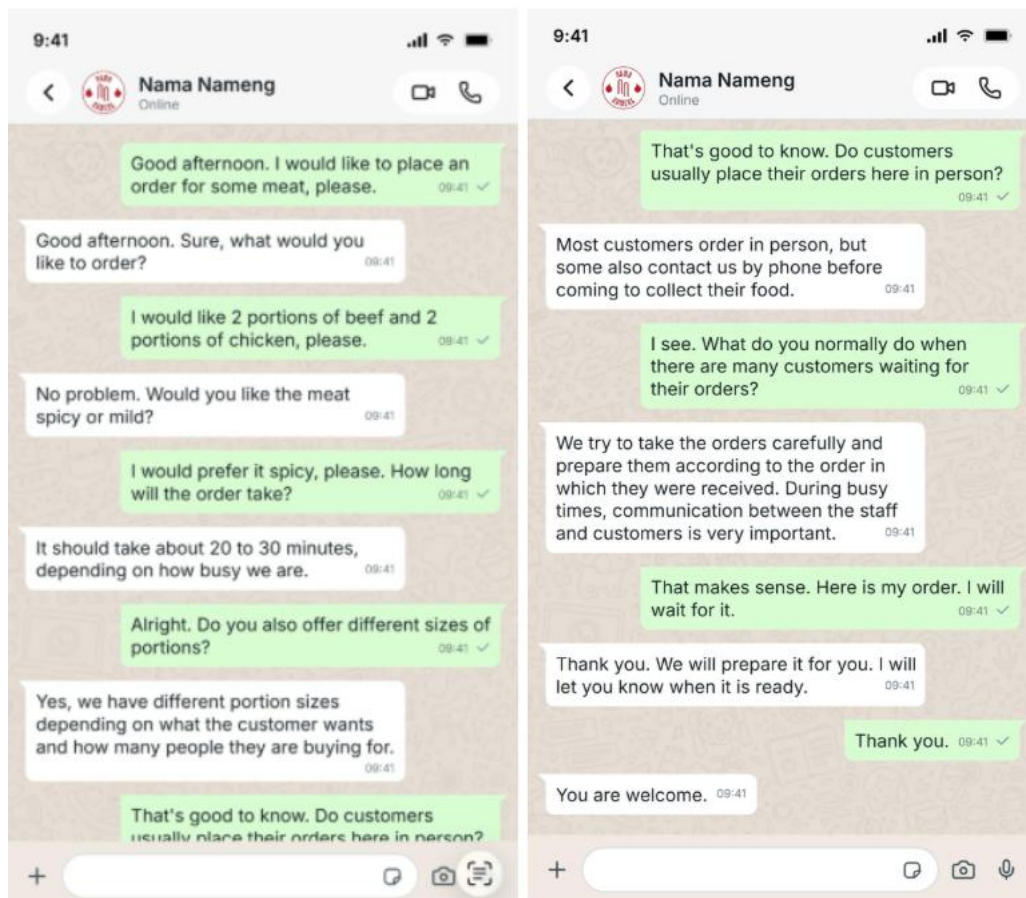
Purpose: Requirements sign-off / review of initial SRS with Doja

Minutes approved by: Charmaine Mogotsi Date: 18/08/2026

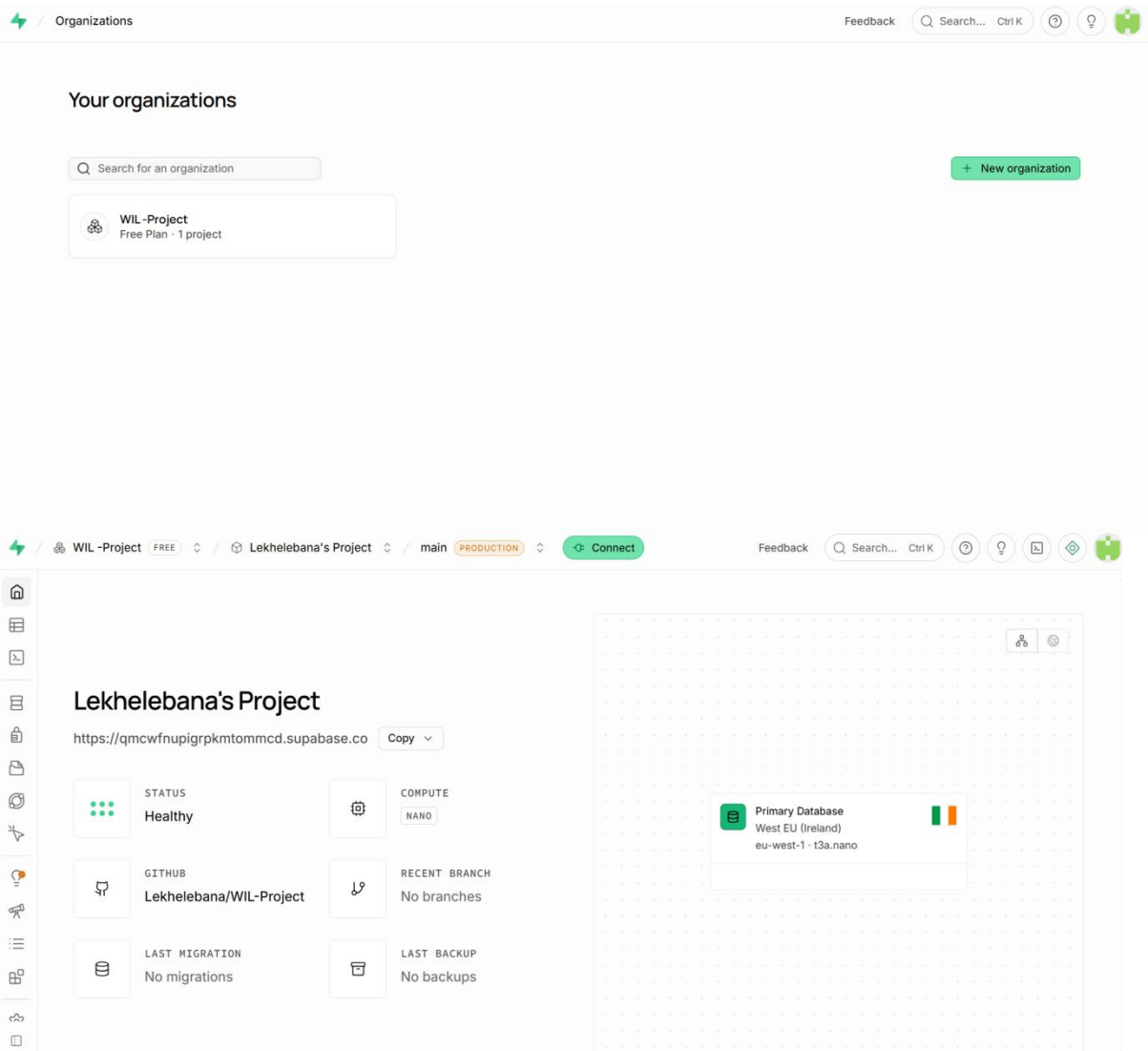
Appendix C: Site Visit / Photographic



Appendix D: Communication Log



Appendix E: Supabase setup



WIL -Project

FREE

Lekhelebana's Project

main

PRODUCTION

Connect

Feedback

Search...

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Settings

CONFIGURATION

General

Infrastructure

Integrations

API Keys

JWT Keys

Log Drains

Add-ons

INTEGRATIONS

Data API

Vault

BILLING

Subscription

Usage

Project Settings

General configuration, domains, ownership, and lifecycle

General settings

Project name

Displayed throughout the dashboard.

Lekhelebana's Project

Project ID

Reference used in APIs and URLs.

qmcwfnpigrpkmtommd

Copy

Project region

West EU (Ireland)

eu-west-1

Copy

Save changes

WIL -Project

FREE

Feedback

Search...

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Team

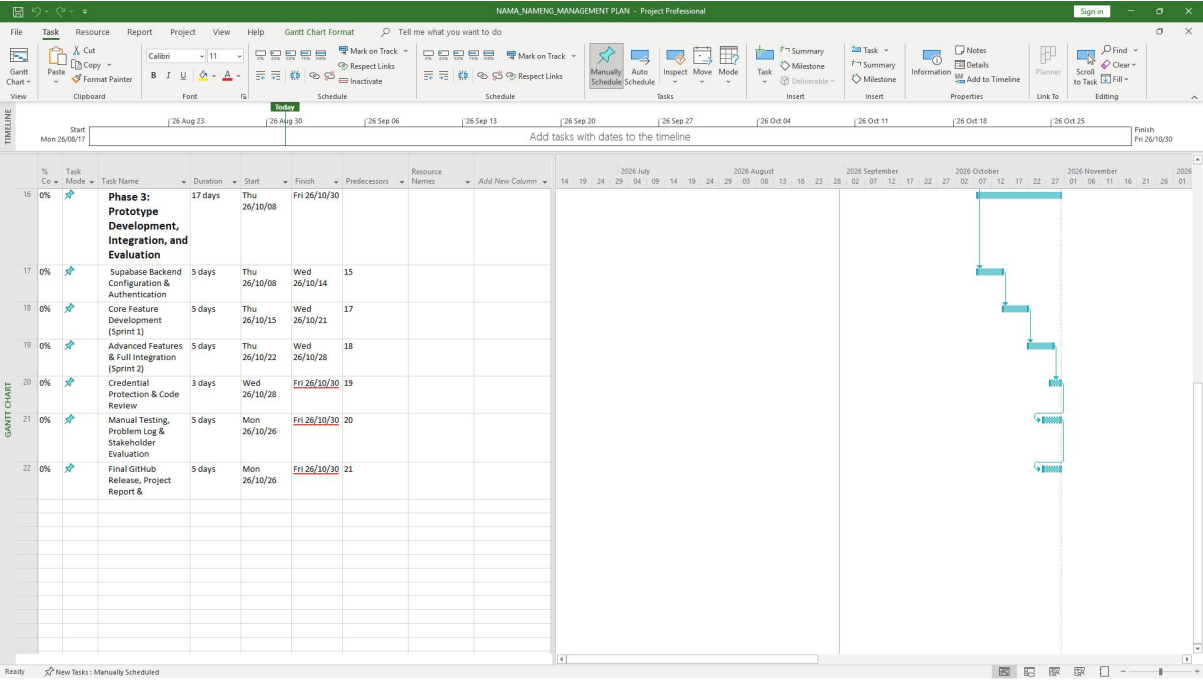
Filter members

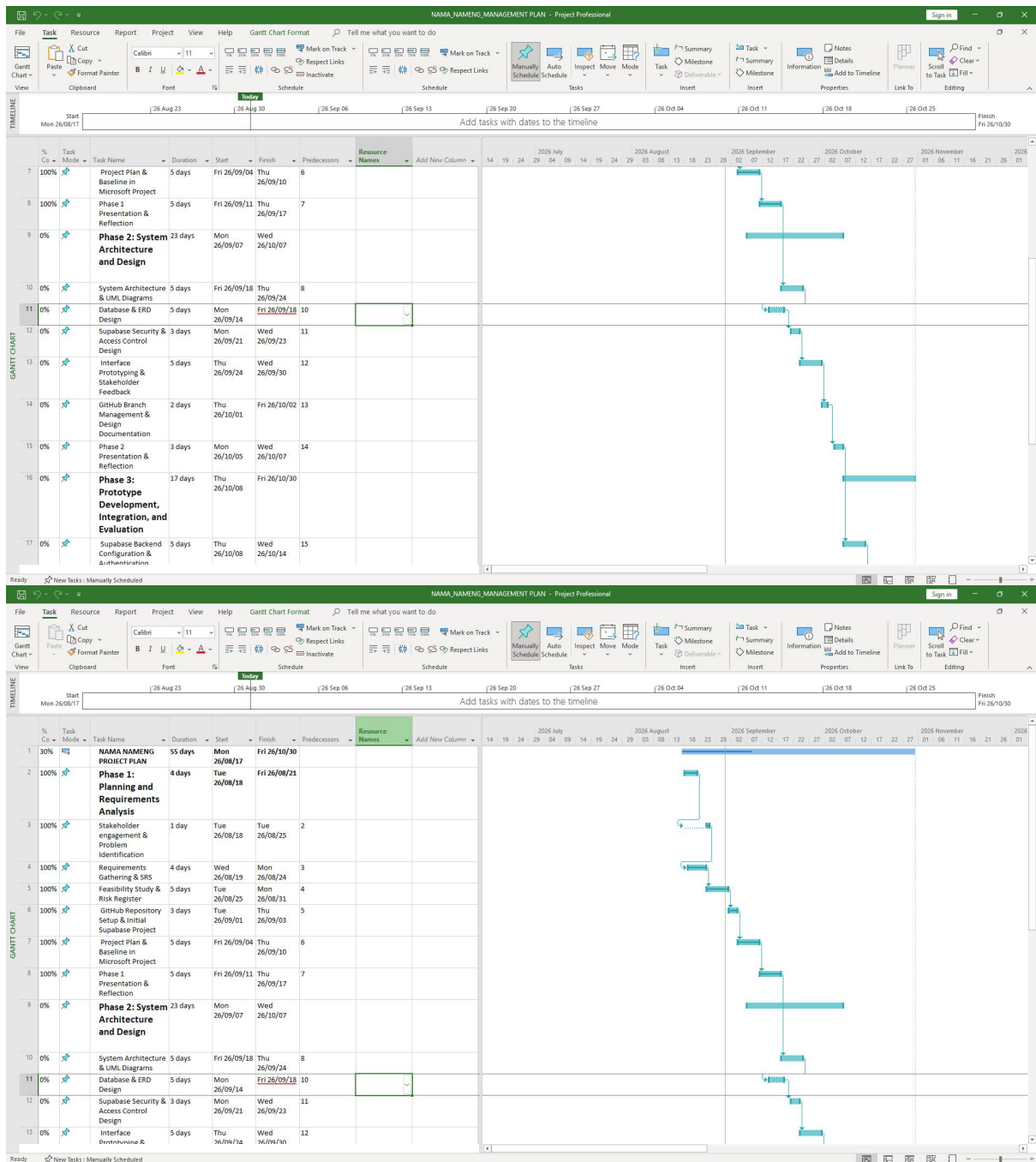
Docs

Invite members

MEMBER	MFA	ROLE	
neolekhelebana12@outlook.com YOU	Disabled X	Owner	Leave team
221027400@stud.cut.ac.za INVITED	Disabled X	Developer	Manage access
mogotsimatumelo@gmail.com INVITED	Disabled X	Developer	Manage access
sefatsaportia17@gmail.com INVITED	Disabled X	Developer	Manage access
tshepangmorake072@gmail.com INVITED	Disabled X	Developer	Manage access

Appendix F: Microsoft Project Plan





Appendix G: Github Repository Link

<https://github.com/Lekhelebana/WIL-Project>